

CORE PLATFORM FRAMEWORK

기술사양서

Public Contract · Starter · Protocol · State · Compatibility

항목	기준
대상	개발자, 아키텍트, QA, 통합 담당자
정본 역할	CPF 기술 계약 Reference
기준 Repository	freeangelsun/202412_01_CPF · master
기준 Commit	b2da6bd720d1a8506db6bddf5d2e35feb9dca964
요구사항 정보	CPF_FINAL_TARGET_REQUIREMENTS.md · 180 개

이 문서는 현재 구현의 상태 보고가 아니라, CPF 제품 사용자가 업무를 끝내기 위해 따라야 할 완성 기준 사용 계약을 중심으로 작성한다. 실제 CPF Symbol/Path 는 CURRENT SOURCE, 구현 독립 예시는 REFERENCE 로 구분한다.

문서 읽는 법과 사실 구분

표기	의미	사용 규칙
CURRENT SOURCE	최신 master 에서 직접 확인한 실제 Class/API/Route/Property/SQL/Test/경로	복사·검색 가능한 실제 이름으로 사용한다.
PRODUCT CONTRACT	CPF 가 완료된 제품으로서 제공해야 하는 사용·운영 계약	현재 구현량 때문에 축소하지 않는다.
REFERENCE	사용자가 그대로 응용할 수 있도록 제공하는 완성 기준 코드·설정·절차 예시	실제 CPF 내부 Symbol 인 것처럼 표현하지 않는다.

본문은 기능을 찾는 데서 끝나지 않는다. 적용 가능한 범위에서 선택 기준 → 준비 → 구현/설정 → 실행 → 정상 결과 → 실패/부분 실패 → 복구·대사 → 보안·감사 → Test → 운영 인계 순서로 설명한다.

1. 기술 Baseline

항목	사양
Java	25
DB	MariaDB/PostgreSQL/Oracle
Frontend	Vue3 + TypeScript + Vite
Batch	Spring Batch Primary Engine
Requirement	180 canonical
Artifact supply	LOCAL_DEV/REMOTE/OFFLINE
Public Profiles	minimal-domain/web-api/secure-api/browser-bff/event-service/batch-service

2. Public Contract 분류

Context

transaction/trace/segment/caller/principal/deadline/attempt

Error

code/protocol mapping/retryable/unknown/field/operator message

Validation

input/output/config/header/file/message/sql bounds

Idempotency

scope/key/requestHash/TTL/result/conflict/concurrency

Locking

lease/fence/owner epoch/version/force release approval

Resilience

timeout/retry/circuit/bulkhead/rate/backpressure

Paging

page/slice/cursor/sort

File

path alias/stream/checksum/atomic publish/security

Message

versioned envelope/correlation/schema/TTL/bounds

Integration

HTTP/TCP/fixed length/provider semantics

Security

principal/authorization/secret/signature/audit

3. Paging 사양

Contract	사양
CpfPageRequest	0-base; default 20; max 200; offset Math.multiplyExact
Sort	direction 기본 ASC; field 는 repository allowlist
Slice	size+1 lookahead
Cursor	stable sort key + signed/versioned/expiring; secret >= 제품 보안 기준

4. Locking 사양

Operation	Result
acquire	ACQUIRED/IDEMPOTENT_REPLAY/BUSY/ RESOURCE_EXHAUSTED/UNKNOWN/INVALID
renew	RENEWED/STALE_TOKEN/NOT_OWNER/NOT_FOUND/EXPIRED/ UNKNOWN/INVALID
release	RELEASED/IDEMPOTENT_REPLAY/STALE_TOKEN/NOT_OWNER/ NOT_FOUND/EXPIRED/UNKNOWN/INVALID
forceRelease	expected fence/epoch/version + approval/command hash

5. Transaction Capability 사양

LOCAL

Canonical state surface: COMMITTED/ROLLED_BACK

XA_JTA

Canonical state surface: ACTIVE/PREPARED/COMMITTED/ROLLED_BACK/HEURISTIC/IN_DOUBT

OUTBOX

Canonical state surface:

PENDING/CLAIMED/PUBLISHED/CONFIRMED/FAILED/UNKNOWN/RETRYING/DEAD/RECONCILED

INBOX_DEDUP

Canonical state surface: RECEIVED/PROCESSING/COMPLETED/FAILED/UNKNOWN/RECONCILED

SAGA

Canonical state surface: STARTED/RUNNING/COMPLETED/FAILED/COMPENSATING/COMPENSATED/UNKNOWN/MANUAL_REVIEW

TCC

Canonical state surface:
TRYING/TRIED/CONFIRMING/CONFIRMED/CANCELING/CANCELED/UNKNOWN/MANUAL_REVIEW

6. Messaging Provider 사양

Provider	구분
Kafka	partition/order/ACK/consumer offset
JMS 3.x	API/runtime abstraction, session/ack/redelivery/transaction
IBM MQ	JMS provider + Queue Manager/Channel/CCDT/TLS/recovery
RabbitMQ	AMQP provider + exchange/queue/ack/redelivery
TCP	persistent connection/framing/heartbeat/reconnect/correlation/response loss

7. Security 사양

- OIDC/OAuth2/JWT: issuer/audience/expiry/claim mapping/login/logout/session/refresh
- KMS/HSM: provider health/key version/rotation/revocation/optional PKCS#11
- Digital Signature: algorithm/keyId/keyVersion/certificate/signature metadata
- Audit: canonical payload/previousHash/currentHash/ordering/concurrency/verification
- Download: reason/approval/row-size/expiry/encryption/watermark/one-time link

8. Gateway 사양

항목	Default/Contract
Port	8070
Connect cap	10s
Response cap	60s
Overall cap	90s
Retry cap	2
Route/policy refresh	30s / 15s
Body cap	10MB req/resp
Control skew/nonce	60s / 180s
Bootstrap	FAIL_CLOSED

9. ADM/BZA Frontend 사양

Route Registry + generated OpenAPI client + typed state/error + Permission/Data Scope + deep link/403/404/session expiry + a11y + Chromium/Firefox/WebKit E2E. Browser storage/URL/DOM/console 에 credential 원문을 노출하지 않는다.

10. Compatibility

Surface	호환성 규칙
Java Public API	semantic version + deprecated replacement
REST/OpenAPI	operationId/DTO/error/header/idempotency 변경 추적
DB	empty install == upgraded final schema manifest
Message	schema version + producer/environment binding
File/FixedLength	layout/version/encoding/checksum
Config	safe default→customer→profile→operation→per-call override
Mixed version	rolling 기간 호환 matrix

11. Starter Value Contract

품질	판정
Convenience	OSS 직접 사용보다 표준 설정/코드 감소
Safe Default	최소 입력으로 안전한 경로
Fail-Fast	Provider conflict/missing secret/unsupported topology 조기 실패
Operations	metric/log/health/audit/runtime control 연결
Recovery	timeout/unknown/reconcile semantics
Escape Hatch	고급 Provider-native 기능을 막지 않음
Zero Footprint	선택하지 않은 capability runtime/Bean 이 남지 않음

12. Reference File Map

Capability	Current Source
Paging	cpf-core/src/main/java/com/cpf/core/api/page/CpfPageRequest.java
Lock	cpf-core/src/main/java/com/cpf/core/api/locking/CpfLockManager.java
Online A-B-C	cpf-reference/src/main/java/com/cpf/reference/online/integrated/OnlineAbcdReferenceFlow.java
HTTP Adapter	cpf-reference/src/main/java/com/cpf/reference/online/integrated/OnlineAbcdReferenceHttpController.java
Spring TX	cpf-reference/src/main/java/com/cpf/reference/online/integrated/OnlineAbcdSpringTransactionService.java
Batch A-B-C	cpf-reference/src/main/java/com/cpf/reference/batch/integrated/BatchAbcReferenceFlow.java
Starter Catalog	cpf-tools/generator/contracts/cpf-starter-catalog.json
ADM Route	cpf-admin/frontend/src/app/routes.ts
BZA Route	cpf-biz-admin/frontend/src/app/routes.ts

Source Trace

구분	Repository 경로	확인 포인트
Canonical	CPF_FINAL_TARGET_REQUIREMENTS.md	requirement-level technical contract
Settings	settings.gradle	build/profile physical contract
Starter Catalog	cpf-starter-catalog.json	leaf/profile/provider

구분	Repository 경로	확인 포인트
Config	cpf-admin/cpf-gateway application.yml	runtime defaults

13. Capability 상세 사양 카드

13.1 Context/TxId

[PRODUCT CONTRACT]

사양 필드	계약
목적	Context/TxId capability 를 제품 Public/Operational contract 로 제공
Public surface	authenticated origin, 34-char txId, segment/parent/attempt, trace/span, deadline
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.2 Error

[PRODUCT CONTRACT]

사양 필드	계약
목적	Error capability 를 제품 Public/Operational contract 로 제공
Public surface	code/http/protocol/retryable/unknown/field/operator
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.3 Validation

[PRODUCT CONTRACT]

사양 필드	계약
목적	Validation capability 를 제품 Public/Operational contract 로 제공
Public surface	size/depth/count/allowlist/field location
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.4 Idempotency

[PRODUCT CONTRACT]

사양 필드	계약
목적	Idempotency capability 를 제품 Public/Operational contract 로 제공
Public surface	scope/key/hash/state/result/ttl/conflict
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.5 Lock

[PRODUCT CONTRACT]

사양 필드	계약
목적	Lock capability 를 제품 Public/Operational contract 로 제공
Public surface	lease/fence/epoch/version/approval recovery
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.6 Resilience

[PRODUCT CONTRACT]

사양 필드	계약
목적	Resilience capability 를 제품 Public/Operational contract 로 제공
Public surface	deadline/timeout/retry/circuit/bulkhead/rate/backpressure
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.7 Paging

[PRODUCT CONTRACT]

사양 필드	계약
목적	Paging capability 를 제품 Public/Operational contract 로 제공
Public surface	page/slice/cursor/sort
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.8 HTTP Client

[PRODUCT CONTRACT]

사양 필드	계약
목적	HTTP Client capability 를 제품 Public/Operational contract 로 제공
Public surface	typed generic response, timeout, auth, trace
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.9 Fixed Length

[PRODUCT CONTRACT]

사양 필드	계약
목적	Fixed Length capability 를 제품 Public/Operational contract 로 제공
Public surface	layout/version/byte length/encoding/padding/group
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.10 File

[PRODUCT CONTRACT]

사양 필드	계약
목적	File capability 를 제품 Public/Operational contract 로 제공
Public surface	stream/checksum/atomic publish/path security
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.11 Attachment

[PRODUCT CONTRACT]

사양 필드	계약
목적	Attachment capability 를 제품 Public/Operational contract 로 제공
Public surface	upload/download/security scan/metadata/retention
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.12 SFTP

[PRODUCT CONTRACT]

사양 필드	계약
목적	SFTP capability 를 제품 Public/Operational contract 로 제공
Public surface	temp target/checksum/rename/status/recovery
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.13 Kafka

[PRODUCT CONTRACT]

사양 필드	계약
목적	Kafka capability 를 제품 Public/Operational contract 로 제공
Public surface	key/partition/order/ACK/consumer commit/rebalance
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.14 JMS

[PRODUCT CONTRACT]

사양 필드	계약
목적	JMS capability 를 제품 Public/Operational contract 로 제공
Public surface	session/ack/redelivery/transaction
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.15 IBM MQ

[PRODUCT CONTRACT]

사양 필드	계약
목적	IBM MQ capability 를 제품 Public/Operational contract 로 제공
Public surface	QM/channel/CCDT/TLS/recovery
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.16 RabbitMQ

[PRODUCT CONTRACT]

사양 필드	계약
목적	RabbitMQ capability 를 제품 Public/Operational contract 로 제공
Public surface	exchange/routing/ack/nack/redelivery
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.17 Outbox

[PRODUCT CONTRACT]

사양 필드	계약
목적	Outbox capability 를 제품 Public/Operational contract 로 제공
Public surface	state/claim/lease/fence/publish/confirm/dead/reconcile
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.18 Inbox

[PRODUCT CONTRACT]

사양 필드	계약
목적	Inbox capability 를 제품 Public/Operational contract 로 제공
Public surface	dedup/processing/completed/unknown/reconcile
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.19 Saga

[PRODUCT CONTRACT]

사양 필드	계약
목적	Saga capability 를 제품 Public/Operational contract 로 제공
Public surface	step/compensation/manual review
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.20 TCC

[PRODUCT CONTRACT]

사양 필드	계약
목적	TCC capability 를 제품 Public/Operational contract 로 제공
Public surface	try/confirm/cancel/expiry/empty rollback
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.21 OAuth2/JWT

[PRODUCT CONTRACT]

사양 필드	계약
목적	OAuth2/JWT capability 를 제품 Public/Operational contract 로 제공
Public surface	issuer/audience/expiry/scopes/claims/security context
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.22 OIDC SSO

[PRODUCT CONTRACT]

사양 필드	계약
목적	OIDC SSO capability 를 제품 Public/Operational contract 로 제공
Public surface	login/logout/session/refresh/idp mapping
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.23 Secret/KMS/HSM

[PRODUCT CONTRACT]

사양 필드	계약
목적	Secret/KMS/HSM capability 를 제품 Public/Operational contract 로 제공
Public surface	provider/key version/rotation/revocation/health
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.24 Digital Signature

[PRODUCT CONTRACT]

사양 필드	계약
목적	Digital Signature capability 를 제품 Public/Operational contract 로 제공
Public surface	algorithm/keyId/version/cert/signature metadata
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.25 Audit

[PRODUCT CONTRACT]

사양 필드	계약
목적	Audit capability 를 제품 Public/Operational contract 로 제공
Public surface	canonical payload/hash chain/signature/verify
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.26 Batch

[PRODUCT CONTRACT]

사양 필드	계약
목적	Batch capability 를 제품 Public/Operational contract 로 제공
Public surface	job/step/chunk/checkpoint/restart/partition
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.27 Center-Cut

[PRODUCT CONTRACT]

사양 필드	계약
목적	Center-Cut capability 를 제품 Public/Operational contract 로 제공
Public surface	target/item/claim/rate/reprocess/unknown
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.28 Gateway

[PRODUCT CONTRACT]

사양 필드	계약
목적	Gateway capability 를 제품 Public/Operational contract 로 제공
Public surface	route/target/security/ledger/publish/LKG
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.29 ADM

[PRODUCT CONTRACT]

사양 필드	계약
목적	ADM capability 를 제품 Public/Operational contract 로 제공
Public surface	route/query/command/permission/approval/recovery
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.30 BZA

[PRODUCT CONTRACT]

사양 필드	계약
목적	BZA capability 를 제품 Public/Operational contract 로 제공
Public surface	organization/role/data scope/approval/session/audit
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

13.31 AI Optional

[PRODUCT CONTRACT]

사양 필드	계약
목적	AI Optional capability 를 제품 Public/Operational contract 로 제공
Public surface	provider/model routing/masking/cost/approval/audit
Configuration	safe default + customer/profile/operation/per-call override 순서
Failure	invalid/timeout/duplicate/partial/unknown/provider failure 를 구분
Security	authorization/masking/secret/audit 적용
Observability	transactionId + capability-specific state/latency/count
Recovery	명시된 retry/restart/reprocess/reconcile/compensation/rollback 만
Compatibility	versioned public contract + migration/deprecation
Verification	actual consumer + negative/fault/runtime scenario

14. 현행 Public Contract Surface

현재 master 에서 새로 확인되는 Public Contract 와 Provider Surface 를 기능 계약에 연결한다.

Capability	Public/Runtime Symbol	계약 핵심	Owner Path
Transaction Strategy	CpfTransactionStrategy	6 strategies	cpf-core/api/transaction
Transaction Definition	CpfTransactionDefinition	propagation/isolation/timeout/readOnly	cpf-core/api/transaction
Transaction Execute	CpfTransactionOperations	provider-neutral execute	cpf-core/api/transaction
TCC	CpfTccParticipant/ CpfTccRecoveryOperations	try/confirm/cancel/recovery	cpf-core/api/transaction
XA	CpfXaTransactionManager/ RecoveryRecord	XA + in-doubt recovery	cpf-core/api/transaction
CRUD	CpfCrudRepositoryPort	generic CRUD + native escape hatch	cpf-core/api/persistence
Search	CpfSearchRepositoryPort/ SearchSpec	filter/search	cpf-core/api/persistence
Persistence Policy	CpfPersistencePolicy	sort/filter/page/bulk guard	cpf-core/api/persistence
AI	CpfAiOperations/Request/Response	transactionId/risk/model/timeout	cpf-core/api/ai
Config Catalog	CpfConfigCatalog/Descriptor/Policy	mutability/config metadata	cpf-core/api/config
Signature	CpfDigitalSignatureOperations	sign/verify	cpf-core/api/security/crypto
Key Lifecycle	CpfKeyManagementOperations/ Provider	key status/rotation provider boundary	cpf-core/api/security/crypto
Tamper Audit	CpfTamperAuditOperations/Store	append/hash-chain storage	cpf-core/api/security/audit
JPA Provider	CpfJpaOperations	timeout/slow-query/bulk/lock/audit	cpf-starters/data/persistence-jpa
JTA Provider	CpfJtaTransactionManagerAdapter	standalone/managed boundary + recovery	cpf-starters/data/transaction-jta
OIDC Provider	CpfOidcPrincipalMapper/ UserService	claim mapping/login	cpf-starters/security/oidc-login
SOAP Provider	CpfSoapClient	SOAP timeout/error client	cpf-starters/integration/soap
AI Provider	CpfAiRouter	provider routing/retry/circuit	cpf-starters/integration/ai
Audit JDBC	JdbcCpfTamperAuditStore	tamper audit JDBC persistence	cpf-starters/security/audit-jdbc